

Blinkit Grocery Sales Analysis: Business Intelligence Report

1. Executive Overview

This project focuses on analyzing Blinkit's grocery sales data to gain insights into sales performance, product demand, outlet efficiency, and customer satisfaction. The analysis combines **Python** for data cleaning and exploratory analysis, **PostgreSQL** for business-oriented SQL analysis, and **Power BI** for developing an interactive dashboard.

The primary objective is to identify key sales trends, evaluate outlet performance, analyze customer purchasing behavior, and generate actionable business insights that can help improve inventory planning, outlet management, and overall business performance.

The project follows a complete analytics workflow, beginning with data preprocessing, followed by SQL-based business analysis, and concluding with an interactive dashboard that enables stakeholders to monitor important business KPIs efficiently.

2. Project Objectives

The main objectives of this project are:

- Analyze Blinkit's grocery sales performance.
- Identify high-performing products and outlet types.
- Evaluate outlet performance across different locations and sizes.
- Measure customer satisfaction using product ratings.
- Build an interactive dashboard for business decision-making.
- Generate data-driven recommendations to improve sales and operational efficiency.

3. Dataset Overview

The dataset contains transactional information about grocery products sold through Blinkit's outlets.

Dataset Summary

Attribute	Details
Total Records	8,523
Total Features	12
Dataset Type	Grocery Sales Dataset

Dataset Features

- Item Identifier
- Item Fat Content
- Item Type
- Outlet Identifier
- Outlet Establishment Year
- Outlet Size
- Outlet Location Type
- Outlet Type

- Item Visibility
- Item Weight
- Total Sales
- Rating

	Item Fat Content	Item Identifier	Item Type	Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size	Outlet Type	Item Visibility	Item Weight	Total Sales	Rating
0	Regular	FDX32	Fruits and Vegetables	2012	OUT049	Tier 1	Medium	Supermarket Type1	0.100014	15.10	145.4786	5.0
1	Low Fat	NCB42	Health and Hygiene	2022	OUT018	Tier 3	Medium	Supermarket Type2	0.008596	11.80	115.3492	5.0
2	Regular	FDR28	Frozen Foods	2010	OUT046	Tier 1	Small	Supermarket Type1	0.025896	13.85	165.0210	5.0
3	Regular	FDL50	Canned	2000	OUT013	Tier 3	High	Supermarket Type1	0.042278	12.15	126.5046	5.0
4	Low Fat	DRI25	Soft Drinks	2015	OUT045	Tier 2	Small	Supermarket Type1	0.033970	19.60	55.1614	5.0
5	low fat	FDS52	Frozen Foods	2020	OUT017	Tier 2	Small	Supermarket Type1	0.005505	8.89	102.4016	5.0
6	Low Fat	NCU05	Health and Hygiene	2011	OUT010	Tier 3	Small	Grocery Store	0.098312	11.80	81.4618	5.0
7	Low Fat	NCB30	Household	2015	OUT045	Tier 2	Small	Supermarket Type1	0.026904	19.70	96.0726	5.0
8	Low Fat	FDW20	Fruits and Vegetables	2000	OUT013	Tier 3	High	Supermarket Type1	0.024129	20.75	124.1730	5.0
9	Low Fat	FDX25	Canned	1998	OUT027	Tier 3	Medium	Supermarket Type3	0.101562	NaN	181.9292	5.0

4. Data Preparation and Preprocessing

Before performing analysis, the dataset was cleaned and prepared using Python.

The following preprocessing steps were performed:

- Imported dataset using Pandas.
- Examined dataset structure using `df.info()`.
- Checked data types.
- Identified missing values.
- Handled missing values in Item Weight.
- Verified duplicate records.
- Standardized categorical values.
- Prepared the cleaned dataset for SQL analysis and Power BI visualization.

```

Item Fat Content      object
Item Identifier       object
Item Type            object
Outlet Establishment Year  int64
Outlet Identifier     object
Outlet Location Type  object
Outlet Size          object
Outlet Type          object
Item Visibility       float64
Item Weight          float64
Total Sales          float64
Rating              float64
dtype: object

```

The cleaned dataset was then exported for further analysis in PostgreSQL.

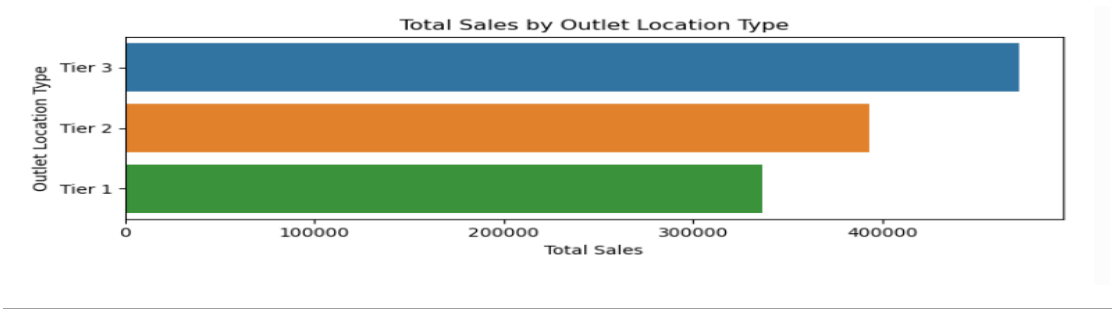
5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the distribution of sales, product categories, outlet characteristics, and customer ratings before conducting business analysis.

Several visualizations were created to identify patterns, trends, and relationships within the data.

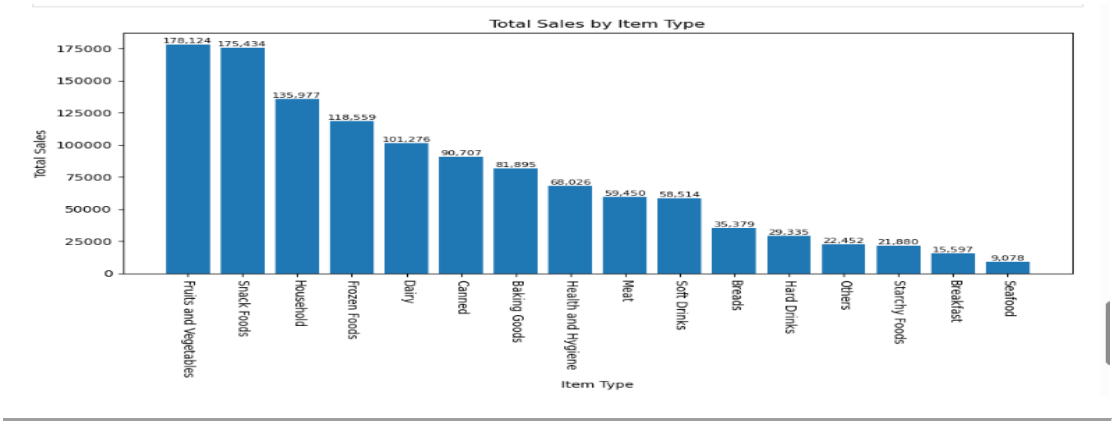
5.1 Sales Distribution

A histogram was created to analyze the distribution of total sales across all products. The visualization shows that sales vary considerably across different products, indicating differences in customer demand.



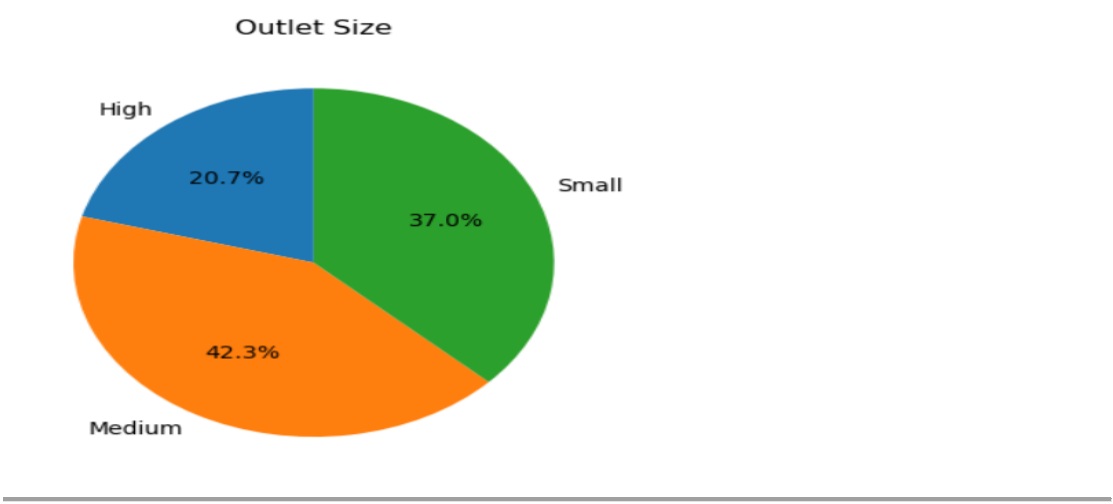
5.2 Item Type Frequency

A count plot was generated to identify the frequency of each product category. This helps determine which product types appear most frequently within the dataset.



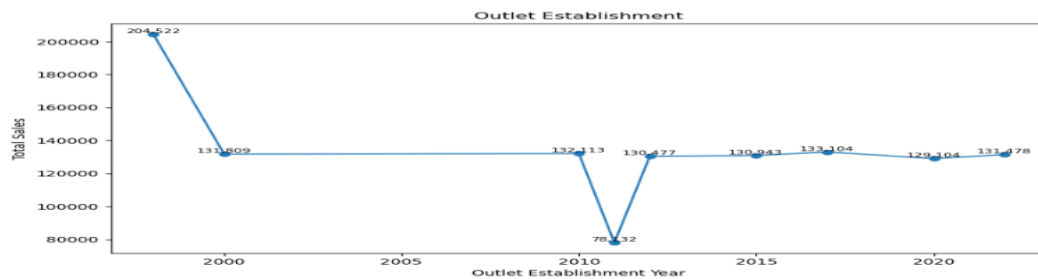
5.3 Outlet Size Distribution

A bar chart was created to analyze the number of outlets classified as Small, Medium, and High. This visualization provides insights into Blinkit's outlet network.



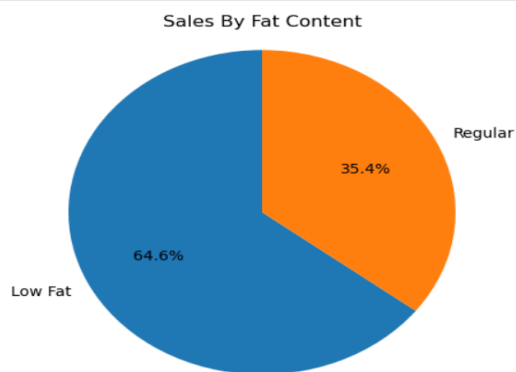
5.4 Outlet Establishment Trend

A visualization was created to understand the distribution of outlet establishment years. This helps identify the company's expansion trend over time.



5.5 Fat Content Analysis

The dataset was analyzed to compare the availability of Low Fat and Regular products.



Key Findings from EDA

- Sales are unevenly distributed across products.
- Several product categories contribute significantly to overall sales.
- Medium-sized outlets represent a large portion of the dataset.
- Blinkit's outlet network has expanded steadily over the years.
- Customer ratings remain consistently positive.
- Low Fat and Regular products both contribute significantly to sales.
- Numerical variables exhibit weak to moderate correlations, indicating that multiple factors influence sales performance.

6. Business Analysis Using PostgreSQL

After completing data cleaning and exploratory analysis, the cleaned dataset was imported into PostgreSQL to perform business-oriented analysis using SQL queries.

The objective was to calculate business KPIs and analyze sales performance across different product categories, outlet types, outlet sizes, and geographical locations.

KPI Analysis

The following Key Performance Indicators (KPIs) were calculated:

- Total Sales
- Average Sales
- Number of Items
- Average Customer Rating

	TOTAL SALES MILLIONS numeric (10,2)		AVERAGE SALES numeric (10)		NUMBER OF ITEMS bigint		AVERAGE RATING numeric (10,2)
1	1.20	1	141	1	8523	1	3.97

Sales by Fat Content

SQL analysis was performed to compare sales generated by Low Fat and Regular products.

	Item Fat Content character varying (80)	TOTAL SALES numeric (10,2)
1	Regular	425361.80
2	Low Fat	776319.68

Sales by Item Type

The total sales generated by each product category were calculated to identify high-performing categories.

	Item Type character varying (50)	TOTAL SALES numeric (10,2)
1	Fruits and Vegetables	178124.08
2	Snack Foods	175433.92
3	Household	135976.53
4	Frozen Foods	118558.88
5	Dairy	101276.46
6	Canned	90706.73
7	Baking Goods	81894.74
8	Health and Hygiene	68025.84
9	Meat	59449.86
10	Soft Drinks	58514.17
11	Breads	35379.12
12	Hard Drinks	29334.68
13	Others	22451.89
14	Starchy Foods	21880.03
15	Breakfast	15596.70
16	Seafood	9077.87

Sales by Outlet Location

Sales were analyzed across different outlet location types to compare geographical performance.

	Outlet Location Type character varying (20) 🔒	Item Fat Content character varying (80) 🔒	TOTAL SALES numeric 🔒
1	Tier 1	Low Fat	215047.91
2	Tier 1	Regular	121349.90
3	Tier 2	Low Fat	254464.77
4	Tier 2	Regular	138685.87
5	Tier 3	Low Fat	306806.99
6	Tier 3	Regular	165326.03

Sales by Outlet Size

Outlet performance was compared based on outlet size to determine which outlet category generated the highest revenue.

	Outlet Location Type character varying (20) 🔒	Low Fat numeric 🔒	Regular numeric 🔒
1	Tier 1	215047.91260	121349.89940
2	Tier 2	254464.77340	138685.86820
3	Tier 3	306806.99240	165326.03480

Sales by Outlet Establishment Year

Sales were analyzed according to outlet establishment year to understand long-term business growth.

	Outlet Establishment Year integer 🔒	TOTAL SALES numeric 🔒
2	2000	131809.02
3	2010	132113.37
4	2011	78131.56
5	2012	130476.86
6	2015	130942.78
7	2017	133103.91
8	2020	129103.96
9	2022	131477.77
Total rows: 9		Query complete 00:00:00.289

Outlet Performance Matrix

A comprehensive analysis was performed for each outlet type including:

- Total Sales
- Sales Percentage
- Average Sales
- Number of Items
- Average Rating

	Outlet Type character varying (30) 🔒	TOTAL SALES numeric 🔒	SALES PERCENTAGE numeric 🔒	AVERAGE SALES numeric 🔒	NUMBER OF ITEMS bigint 🔒	AVERAGE RATING numeric 🔒
1	Supermarket Type1	787549.89	65.54	141.21	5577	3.96
2	Grocery Store	151939.15	12.64	140.29	1083	3.99
3	Supermarket Type2	131477.77	10.94	141.68	928	3.97
4	Supermarket Type3	130714.67	10.88	139.80	935	3.95

7. Interactive Dashboard (Power BI)

An interactive Power BI dashboard was developed to visualize business performance and provide meaningful insights through dynamic visualizations.

The dashboard includes:

KPI Cards

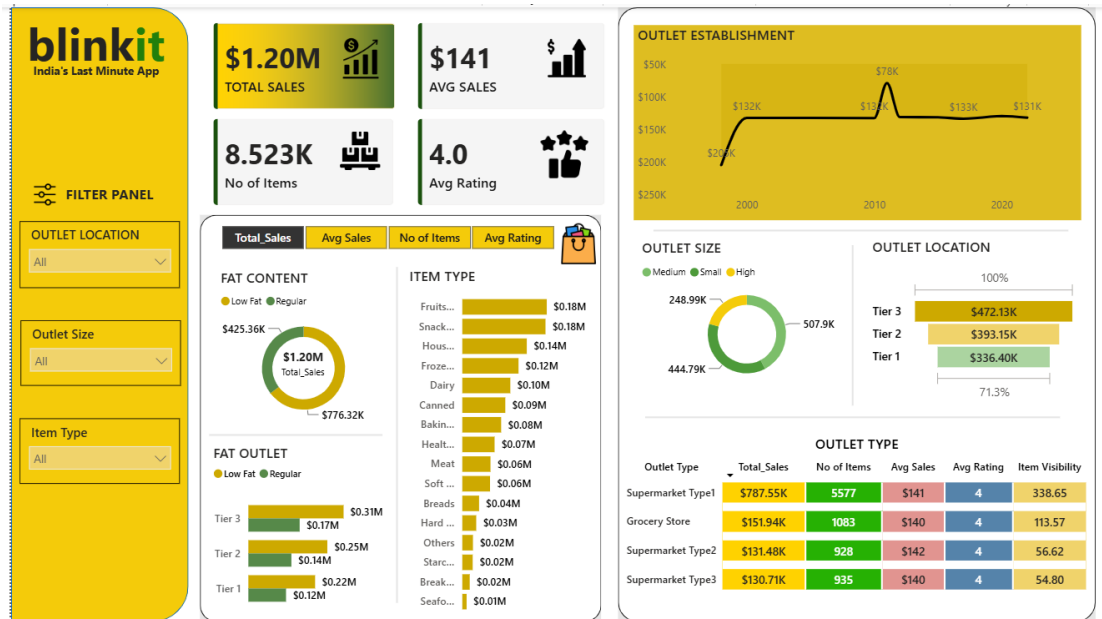
- Total Sales
- Average Sales
- Number of Items
- Average Rating

Charts

- Sales by Outlet Establishment Year
- Sales by Item Type
- Sales by Fat Content
- Sales by Outlet Size
- Sales by Outlet Location
- Fat Content by Outlet
- Outlet Performance Matrix

Interactive Features

- Dynamic slicers
- Cross-filtering
- Reset filter button
- Navigation buttons
- Interactive visual exploration



8. Key Business Insights

The analysis revealed several valuable business insights:

- Medium-sized outlets generated the highest overall sales.
- Supermarket Type1 emerged as the highest-performing outlet type.
- Tier 3 outlet locations contributed significantly to total revenue.
- Fruits & Vegetables and Snack Foods were among the highest-selling product categories.
- Both Low Fat and Regular products demonstrated strong customer demand.
- Customer ratings remained consistently high, indicating positive customer satisfaction.
- Recently established outlets showed encouraging sales performance, reflecting Blinkit's business expansion.

9. Business Recommendations

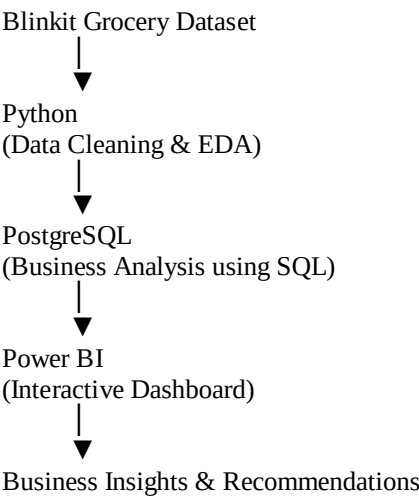
Based on the findings, the following recommendations are proposed:

- Increase inventory for high-performing product categories.
- Focus expansion efforts on high-performing outlet locations.
- Optimize inventory allocation based on outlet size and demand.
- Improve the performance of lower-performing outlet types through targeted promotions.
- Continue monitoring customer ratings to enhance service quality.
- Utilize interactive dashboards for continuous business performance monitoring and strategic decision-making.

10. Technologies Used

Technology	Purpose
Python	Data Cleaning & EDA
Pandas	Data Manipulation
NumPy	Numerical Computing
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
PostgreSQL	SQL-Based Business Analysis
pgAdmin 4	Database Management
Power BI	Dashboard Development
Git & GitHub	Version Control & Project Hosting

11. Project Workflow



12. Conclusion

This project demonstrates a complete end-to-end Business Intelligence workflow by integrating Python, PostgreSQL, and Power BI to analyze Blinkit's grocery sales data. Through data cleaning, exploratory analysis, SQL-based business analysis, and interactive dashboard development, the project successfully identified sales trends, outlet performance, customer preferences, and operational insights. The generated recommendations can support better inventory planning, outlet optimization, and strategic business decision-making. The project showcases practical skills in data analytics, SQL querying, business intelligence, and dashboard design, making it a strong portfolio project for aspiring Data Analysts.